Avinash Agarwal

Mathematics | Engineering | Artificial Intelligence | Distributed Systems

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The United States of America

Summary

Curiosity-driven engineering veteran with 8 years of industry-leading experience at Meta and Microsoft; specializing in full-stack development, distributed cloud infrastructure, and high-throughput data-pipelines. Proven track record of designing and managing high-volume distributed systems processing billions of daily notifications. Passionate about leveraging Artificial Intelligence and Distributed Systems to solve complex planet-scale engineering challenges.

Experience

Meta

August 2025 — Present

Engineering Lead

Menlo Park, California, USA

- **Architecting Global Identity Systems:** Ensuring high-availability and security for identity systems processing billions of requests across the entire Meta ecosystem.
- **Secured runners-up position in an internal Hackathon**, demonstrating high engineering velocity by rapidly triaging and fixing a high volume of bugs.
- **Contributed to the development of a company-wide automated on-call agent**, significantly reducing manual triage time and improving engineering team productivity."
- **AI for Longevity at Meta Superintelligence Labs:** Spearheading technical advocacy and research to improve AI powered Health Benchmarks.

Microsoft Azure

Feb 2022 — July 2025

Engineering Lead

Redmond, Washington, USA

Core Engineering | Azure Resource Graph

- **Root-caused and advocated for increased hash cardinality** across the storage stack, leveraging data-driven demonstrations to significantly improve distribution granularity and resolve partition skewness.
- **High-Performance Serialization:** Evaluated and advocated for the adoption of **System.Text.Json** across the service stack, leveraging its high-performance, low-allocation APIs to achieve significant reductions in CPU overhead and memory pressure during large-scale telemetry processing.
- **Engineered data ingestion components** hosted on the Azure Synapse platform to support dynamic time-window loading for security datasets.
- Optimized complex SQL queries executed on Azure Synapse to efficiently load and merge multiple continuous data streams.
- Worked on migrating a streaming data pipeline (1B+ daily notifications) to its dedicated scale unit — Involved sophisticated design, careful cross-team planning and execution.
- Profiled and optimized high-contention services eliminating bottlenecks and achieving **nearly 10%** improvement in compute, memory and network efficiency.
- Improved on-call efficiency with multiple improvements to internal monitors, documentations, etc.

Engineering Lead | Azure Resource Change Analysis

- Delivered General Availability for the Change Analysis back-end by successfully deploying the service to US and China Government Sovereign Clouds, ahead of an important Gartner cloud review.
- Significantly reduced partition skewness in data-ingestion pipelines, achieving **over 30%** improvement in compute, memory and network efficiency.
- Led the architectural evolution, including metadata enhancements and optimizations to the Change Analysis payload with extensive collaboration across teams and disciplines.
- Delivered General Availability for the Change Actor functionality, enabling customers to audit **WHO** initiated a change and **WHICH client** (Portal, CLI, SDK) was used to make these changes.
- Delivered General Availability for the Change Analysis front-end (live on the Azure portal) by improving performance Lighthouse scores, automating deployments and tests, completing accessibility fixes etc.

Azure Global, Microsoft — *June 2018 — Jan 2022*
Engineering Lead — Hyderabad, Telangana, India

BCDR & Global Traffic Management: Engineered a mission-critical Business Continuity and Disaster Recovery architecture using Azure Front Door, establishing a resilient global load-balancing layer slashing page-load latencies and significantly hardening the platform's security posture for massive-scale international traffic.

Scaling for GRAB (Strategic Partnership)

- **Performance Profiling & Load Testing:** Collaborated with a specialist engineering partner to perform intensive load testing and performance profiling, ensuring platform stability for massive-scale deployment across Southeast Asia.
- **System Optimization & Throughput:** Optimized core services to achieve a **20x throughput improvement** by studying and refining SQL query execution plans, eliminating redundant network calls, and implementing strategic caching layers.
- **High-Scale Data Migration:** Architected APIs to execute the **migration of 160M+ course-completion records** from AWS to Azure, designing robust APIs to ensure zero-downtime and data integrity during the transition.
- **Technical Leadership & Design:** Spearheaded technical design reviews and PM alignment, advocating for extensible solutions to avoid technical debt from short-term fixes.

UNICEF Learning Passport (powered by Microsoft Community Training)

- **Offline Innovation (Hackathon Lead):** Spearheaded a hackathon project to host the entire cloud service locally on Windows machines. The platform was recognized as one of **TIME's Best Inventions of 2021** for providing continuous education to children in conflict zones and refugee settings.
- **Global Societal Impact:** This "offline-first" prototype was the key technical catalyst that prompted **UNICEF** to launch the initiative using Microsoft Community Training for no-connectivity regions, contributing to a high-scale initiative that reached **7.4 million learners across 43 countries** by 2024;
- **Strategic Architectural Leadership:** Successfully advocated for, and worked on containerization and migration to **cross-platform .NET Core**; this move was critical for the platform's ability to run on diverse hardware in emerging markets.
- **Operational Support & Edge Strategy:** Facilitated the integration of **Azure IoT Hub** to support remote management and observability for offline learning hubs; assisted in testing data-sync on diverse Linux-based hardware to ensure reliable curriculum delivery in remote field operations.

Swachh Bharat Mission (powered by Project Sangam)

- Directed an engineering team through the end-to-end implementation of key mobile features, culminating in the successful launch of the official Swachh Bharat Mission eLearning mobile app.
- Spearheaded the rapid triage of launch-critical issues in close collaboration with Aditya Agarwal and Product Management teams to ensure a smooth migration of the training platform.
- Architected and implemented a high-impact search-enabled dropdown to resolve a critical UI bug; this allowed users to efficiently navigate and select from **the hundreds of cities where the platform was deployed**.
- **Contributed to a high-scale social impact project** that successfully trained and empowered over **110,000+ municipal functionaries** across **4,000+ cities** in India.

Platform Localization

- Spearheaded global accessibility by **engineering an internal localization framework** and integrating with manual translation pipelines.
- Initially developed comprehensive support for **right-to-left (RTL) formats** before assuming full platform localization ownership.
- Conducted a **comprehensive investigation on ligature support** to ensure accurate rendering of complex scripts across the ecosystem.
- **Custom PDF Certification Engine:** Architected a templatized HTML-to-PDF generation feature by defining rigorous localization and security requirements; strategically collaborated with IronPDF to accelerate delivery while ensuring full support for complex scripts and global accessibility standards.

Notifications Infrastructure

- Initially onboarded via Xamarin consumption APIs before assuming full ownership of the platform's notification subsystem, under the mentorship of Rohan Badlani.

- **High-Throughput Bulk Ingestion:** Architected and built an end-to-end bulk upload system enabling parallel ingestion of multi-format media; significantly reduced administrative onboarding time and **super-charged the manual testing experience** for everyone.

- **Developed a scalable monitoring infrastructure** using Event Hubs and Azure Data Explorer to centralize telemetry and enable diagnostic and analytic dashboards.
- **Automated end-to-end deployments and CI/CD pipelines**, enabling the Public Preview release and publishing of the final offering on the Azure Marketplace.
- **Identity & Authentication:** Streamlined the SMS-based login flow by implementing custom Identity Server solutions over Azure Cosmos DB and closely collaborated with Telesign to significantly reduce SMS code-sending costs across many countries.
- **Search Infrastructure & Optimization:** Designed and implemented a custom search index featuring advanced schema optimizations and Lucene-based retrieval.
  - Integrated with custom skills to extract metadata from PDFs, videos, and documents into indexed columns.
  - Enhanced discovery by implementing **fuzzy matching, synonym mapping, and phonetic analyzers** using Azure Cognitive Search to improve query relevance.
- **Media Security & DRM:** Authored a technical proposal to modernize content protection for temporal media using industry-standard **DRM (Digital Rights Management)** including **Widevine, PlayReady, and FairPlay** over existing/legacy AES-128 encryption.
- **Advanced Thread Modeling:** Conducted deep-dive threat modeling and attack surface analysis using Microsoft internal security frameworks; ensuring the integrity of user data and platform content.
- **Privacy & Compliance:** Enabled automated compliance tooling to automate the identification of secure, licensed dependencies; ensuring rigorous adherence to global data protection and privacy standards.
- **State Persistence & User Analytics:** Developed features for real-time tracking of user progress, including "most recently viewed lesson" and "last watched timestamp" for temporal media content ensuring a seamless cross-device learning experience.
- **Delivered end-to-end features and bug fixes** spanning data layers, backend services, and frontend components.

Microsoft Hackathon

June — 2018

Engineer — Hyderabad, Telangana, India

- **Kool AI(d) for Gamers:** Extracted live game metrics and game states from DotA-2 streams for real-time analytics.

Microsoft Code.Fun.Do

2018 — 2019

Engineering Lead and Logistics Moderator — India

- **Competitive Programming Strategy:** Strongly advocated for the adoption of structured competitive programming contests (to survive against modern industry leaders like Google and Meta.)
- **Event Moderator:** Delivered technical talks and managed event logistics at **BITS Pilani, Hyderabad** and **NIT Surathkal**.

The Garage, Microsoft

June — 2018

Engineer — Hyderabad, Telangana, India

- **IoT Home Automation:** Led a team of 5 to work on retrofitting legacy electric switches with motorized 3D-printed actuators and control software.
- **Assembling Bicycles:** Social volunteering opportunity, as part of Microsoft's Giving Campaign.

Microsoft Foundry

May 2016 — July 2016

Engineering Internship — Hyderabad, Telangana, India

- **Conversational AI:** Engineered chat-bots using the **Microsoft Bot Framework**, architecting custom conversational flows for automated form-filling.
- **Collaborative Development:** Contributed to a high-impact project within a cross-functional team of **8 members**.
- **Computer Vision & NLP:** Helped enhance OCR performance using **OpenCV** and test refined natural language experiences using **Microsoft LUIS**.

Xerox Research Centre, India

Dec 2015 — Jan 2016

Machine Learning Internship — Bengaluru, Karnataka, India

- **Xerox Research Innovation Challenge:** Engineered predictive neural network models using vitals and lab data to forecast patient mortality and health complications among ICU patients.
- Secured **3rd place** (online) and **7th place** (onsite) in the challenge alongside a 3-person team, successfully competing against advanced Ph.D. research teams.
- Completed an intensive Winter School on Machine Learning under the mentorship of Vaibhav Rajan.

Projects

- **AIMA-CSharp Open Source Contribution:** Contributed to the AIMA-CSharp repository by implementing pseudo-code in C# from Peter Norvig's *Artificial Intelligence: A Modern Approach*.
- **Image Steganography:** Implemented a system to encrypt and embed hidden messages within images.
- **Multi-Agent Environment Simulation:** Built a 2D grid simulation of agents coordinating with one another.
- **2D Physics Game Engine:** Developed a custom rigid-body and gravitational physics engine in Java focusing on real-time kinematic interactions.
  - Implemented efficient **collision detection and resolution** for multiple dynamic objects.
  - Engineered an **environmental boundary system** to manage object interactions with surrounding walls.
  - Implemented a **Near-Earth constant gravity model** to simulate realistic downward acceleration.
  - Implemented a **Newtonian** $1/r^2$ **gravity model** to simulate object interaction in space.
- **Intel E5200 CPU Overclocking**: Successfully overclocked an Intel Pentium E5200 processor from a base clock of **2.5GHz to 2.8GHz** to improve gaming performance on budget hardware.
  - Optimized system performance by manually adjusting **FSB frequencies, CPU multipliers, and core voltage settings** within the BIOS to achieve a stable 12% frequency increase
  - Conducted extensive stress testing to monitor thermal output and system stability under peak loads, ensuring long-term hardware reliability and optimal heat dissipation
  - Demonstrated system stability and tangible FPS improvements on the PC port of Grand Theft Auto IV.
- **Tic-Tac-Toe AI:** Implemented a decision-tree based AI and multi-player for Tic-Tac-Toe.

Education

Indian Institute of Technology - Varanasi
July 2013 - May 2018

Integrated Master's degree (5 years): Mathematics & Computing
Varanasi, India

- **Thesis:** Deep Learning Based Classification of Handwritten Mathematical Symbols
  (Supervised by Tanmoy Som)
  - Trained transfer learning models achieving state-of-the-art accuracy on the HasyV2 dataset.
- **Teaching Assistant** for the freshman course on **Mathematical Methods**
  (Supervised by Manushi Gupta)
- **Problem Setter & Moderator – Hackerrank (Technex'16):**
  - **Mathletics:** www.hackerrank.com/mathletics-technex16
  - **Prayaas:** www.hackerrank.com/prayaas-technex16
- Ranked 1st in CodeDef 2016, a coding contest organized in Udyam, a college festival of IIT Varanasi.
- Ranked 2nd in Analiticity, a campus-wide Data Science competition at Technex '16.
- Ranked 1st in COPS-OPC-2015, a programming contest organized by the club of programmers at IIT Varanasi.
- **Microsoft Code.Fun.Do** Developed a 2D top-down space shooter game using the Unity game engine.
- **Predicting Tourism Trends:** Participated in a team of 3 to build machine learning models to forecast tourism trends (Supervised by Santwana Mukhopadhyay)
- **Rediscovering Darkness:** Runner's up in the Astronomy Club event
- **Treasure Hunt:** Runner's up in a team of 3, a social event organized under KashiYatra
- **Event Organizer & Moderator**: Moderated competitive AoE-2, CoD-4 and DotA-2 tournaments.
- **Robo-Soccer:** Runners-up in a team of 5, at a Robo-Soccer event organized by the Robotics Club
- **The Chess Game:** Developed an evolving chess engine (team of 2) from pre-college through my final year, transitioning from a foundation in Java to a sophisticated AI-driven system.
  - **Phase 1:** Engineered real-time multiplayer functionality during my first year.
  - **Phase 2:** Independently integrated an AI engine utilizing **Mini-Max** and **Alpha-Beta pruning** to optimize decision-tree search.

- **DeepLearning.AI:** Deep Learning Specialization — *August 2017*
- **University of Alberta & Amii:** Reinforcement Learning Specialization — *July 2020*

- **The State University of New York:** Blockchain Basics *June 2021*
- **DeepLearning.AI:** Generative AI For Software Development *February 2025*
- **The Lex Fridman Podcast**: AI, Blockchain, Science, Mortality and Philosophy *2019 — Present*
- **The Jordan B Peterson Podcast**: Maps of Meaning, Clinical Psychology, Philosophy *2018 — Present*

ACCOMPLISHMENTS

- **Competitive Programming, Machine Learning and Quant**
  - Qualified for and represented IIT Varanasi at **ACM-ICPC 2016 Regionals**.
  - Qualified for and represented IIT Varanasi at **ACM-ICPC 2015 Regionals**.
  - Ranked 15th worldwide (2nd in India) in **Codeforces Round #410 (Div. 2)**.
  - **GS Quantify 2016:** National Finalist (Top 2 in India); Led a team of 3 in Goldman Sachs' flagship challenge. Managed the end-to-end execution of algorithmic trading and ML tracks; Helped with feature engineering for Bond Liquidity prediction models; spearheaded derivative-based quantitative analysis.
  - **IndiaHacks 2016:** Collaborated with **Sartaj Singh**, officially securing an **overall 14th rank** against 3,000+ other teams in the Machine Learning track; Received official invitation to the onsite finals at **Vivanta Taj, Bangalore**.
  - Ranked 83rd worldwide in **Google Kickstart** Round D 2017.

- **Academic**
  - All India Rank 494 in Joint Entrance Examination - JEE (MAINS) - 2013, among 1.3 million candidates.
  - All India Rank 1672 in IIT-JEE (ADVANCED) - 2013, among 150k candidates.
  - All India Rank 6355 in National Eligibility cum Entrance Tests (NEET) - 2013, among 650k candidates.
  - Scored 96.8% in the All India Senior School Certificate Examination - 2013.
  - Awarded DST INSPIRE Science Scholarship for ranking in the top 1% of AISSCE 2013.
  - Recognized in the top 0.1% of AISSCE 2013 for excellence in Physical Education.
  - Achieved Rank 1 in Medical stream of Jharkhand Combined Entrance Competitive Examination - 2013.
  - Achieved Rank 2 in Engineering stream of Jharkhand Combined Entrance Competitive Examination - 2013.

SKILLS

- **Engineering Leadership:** Technical Design Reviews, Mentorship & Team Leading, On-call Engineering & Incident Response, Security & Privacy Reviews, Threat Modeling, Cross-functional Collaboration
- **AI / Machine Learning:** LLM Applications & Agentic Workflows, Reinforcement Learning, Deep Learning (CNNs, RNNs, Transfer Learning etc.), Machine Learning (Statistics, Classification, Regression, SVMs, PCA etc.)
- **Backend & Distributed Systems:** BCDR (Business Continuity & Disaster Recovery), High-Scale Services (Billions of daily requests), Distributed Systems Architecture, API Design, Data Pipelines, Authentication & Identity Systems, Blockchain.
- **Data & Storage:** Blockchain, NoSQL/SQL, Redis, Search Infrastructure (Lucene/Elasticsearch)
- **Cloud & Infrastructure:** Microsoft Azure (Expert), AWS, Kubernetes, CDN, CI/CD & Infrastructure Automation
- **Frameworks & Libraries:** .NET / ASP.NET Core, React, TensorFlow, Keras, Pandas, NumPy, Scikit-learn, XGBoost, OpenCV, Unity, IronPDF
- **Programming Languages:** C, C++, C#, Java, Python, TypeScript, Scala, Hack/PHP, SQL, PowerShell, Octave
- **Natural Languages:** English and Hindi

INTERESTS

- Science Fiction — Video-games, Podcasts, documentaries, books, etc.
- Music — Pop and Alternative Rock.
- Exploration — Future expeditions to Alaska and Hawaii!
- Comedy — Bill Maher, Stephen Colbert
- Guitar — Beginner
- Cooking — Beginner
- Badminton — Beginner